

In Vivo Validation of Modulated Acoustic Radiation Force–Based Imaging in Murine Model of Abdominal Aortic Aneurysm Using VEGFR-2–Targeted Microbubbles

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Objectives: The objective of this study is to validate the modulated acoustic radiation force (mARF)–based imaging method in the detection of abdominal aortic aneurysm (AAA) in murine models using vascular endothelial growth factor receptor 2 (VEGFR-2)–targeted microbubbles (MBs).

Materials and Methods: The mouse AAA model was prepared using the subcutaneous angiotensin II (Ang II) infusion combined with the β -aminopropionitrile monofumarate solution dissolved in drinking water. The ultrasound imaging session was performed at 7 days, 14 days, 21 days, and 28 days after the osmotic pump implantation. For each imaging session, 10 C57BL/6 mice were implanted with Ang II–filled osmotic pumps, and 5 C57BL/6 mice received saline infusion only as the control group. Biotinylated lipid MBs conjugated to either anti–mouse VEGFR-2 antibody (targeted MBs) or isotype control antibody (control MBs) were prepared before each imaging session and were injected into mice via tail vein catheter. Two separate transducers were colocated to image the AAA and apply ARF to translate MBs simultaneously. After each imaging session, tissue was harvested and the aortas were used for VEGFR-2 immunostaining analysis. From the collected ultrasound image data, the signal magnitude response of the adherent targeted MBs was analyzed, and a parameter, residual-to-saturation ratio ($R_{res-sat}$), was defined to measure the enhancement in the adherent targeted MBs signal after the cessation of ARF compared with the initial signal intensity. Statistical analysis was performed with the Welch *t* test and analysis of variance test.

Results: The $R_{res-sat}$ of abdominal aortic segments from Ang II–challenged mice was significantly higher compared with that in the saline-infused control group ($P < 0.001$) at all 4 time points after osmotic pump implantation (1 week to 4 weeks). In control mice, the $R_{res-sat}$ values were 2.13%, 1.85%, 3.26%, and 4.85% at 1, 2, 3, and 4 weeks postimplantation, respectively. In stark contrast, the $R_{res-sat}$ values for the mice with Ang II–induced AAA lesions were 9.20%, 20.6%, 22.7%, and 31.8%, respectively. It is worth noting that there was a significant difference between the $R_{res-sat}$ for Ang II–infused mice at all 4 time points ($P < 0.005$), a finding not present in the saline-infused mice. Immunostaining

results revealed the VEGFR-2 expression was increased in the abdominal aortic segments of Ang II–infused mice compared with the control group.

Conclusions: The mARF-based imaging technique was validated in vivo using a murine model of AAA and VEGFR-2–targeted MBs. Results in this study indicated that the mARF-based imaging technique has the ability to detect and assess AAA growth at early stages based on the signal intensity of adherent targeted MBs, which is correlated with the expression level of the desired molecular biomarker. The results may suggest, in very long term, a pathway toward eventual clinical implementation for an ultrasound molecular imaging–based approach to AAA risk assessment in asymptomatic patients.

Key Words: acoustic radiation force, microbubbles, ultrasound molecular imaging, abdominal aortic aneurysm, vascular endothelial growth factor receptor

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An abdominal aortic aneurysm (AAA) is defined as a segmental, full-thickness dilation of the abdominal aorta exceeding the normal vessel diameter by 50%.¹ Most AAAs are silent and asymptomatic until rupture.^{1,2} However, rupture is often lethal with an in-hospital survival rate just above 50%.¹ Aortic aneurysms result in approximately 15,000 deaths per year in the United States,^{3,4} and it is estimated to lead to 150,000–200,000 deaths each year worldwide.^{5,6}

Aneurysm diameter is currently one of the most used predictors of the risk of AAA rupture. An aortic diameter ≥ 3.0 cm is commonly used as the ultrasonic criterion for AAA,^{7,8} and an aneurysm diameter of 5.5 cm is usually considered the threshold for surgery repair intervention.⁹ However, AAA is a multifactorial disease, and several individual characteristics may influence aneurysm growth in diameter,¹⁰ such as cigarette smoking,¹¹ diabetes,¹¹ and gender,¹² and so on. The surgical intervention threshold may need to be justified given the specific risk factors of individual patients.¹¹ Therefore, several AAA detection techniques, as alternatives to routine ultrasonography screening,¹³ have been developed recently. Gandhi et al¹⁴ investigated the feasibility of using fluorine-18-fluorothymidine, the cell proliferation radiotracer, with positron emission tomography/computed tomography as an early marker in a progressive AAA model. Yao et al¹⁵ reported a novel method using activatable nanoprobe combined with magnetic resonance imaging to detect the matrix metalloproteinases activity in the early stages of AAA. Dong et al¹⁶ introduced and evaluated a technique based on aortic magnetic resonance elastography derived AAA stiffness to assess AAA wall status and rupture potential using non-contrast-enhanced magnetic resonance imaging. These AAA detection techniques were anticipated to detect and assess the AAA progression at early stages, as well as provide personalized patient monitoring to evaluate the necessity for early surgical intervention.¹⁴

Ultrasound molecular imaging (USMI), which characterizes and measures biological processes at the cellular and molecular level, is a promising technique with the potential to detect and diagnose AAA.^{17–19} It benefits from the traditional, relatively low-cost, ultrasound diagnostic imaging instrumentation and uses molecular probes, in the form of targeted contrast agents, to highlight the existence of molecular markers that indicate the disease of interest.^{20,21}

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Ultrasound contrast agents generally comprise gas-filled microbubbles (MBs).^{22–24} Microbubbles can be designed to image physiological processes on a molecular scale by decorating with specific ligands that possess affinity toward the molecular markers associated with the diseases of interest such as atherosclerosis,^{25,26} cancer,^{27–29} and inflammation.^{30,31} The inflammatory cell infiltration and the hyperactivation of proteolytic enzymes are involved in the aneurysm formation and expansion.^{32,33} These biological processes are believed to result in the destruction of the aortic media and vascular smooth muscle cell apoptosis and dysfunction as a result of the release of a range of proteolytic enzymes, free radicals, cytokines, and related products.⁵ Multiple biomarkers, including vascular endothelial growth factor A (VEGF-A) and its receptors, may contribute to the progression of AAAs.³⁴ Tedesco et al³⁵ found that VEGF receptor (VEGFR) expression, as determined by fluorescent imaging and immunostaining, increased in experimental AAAs in a diameter-dependent fashion. Consequently, as an alternative for detecting and assessing the features of AAA, USMI attempts to delineate the AAA lesions by selectively visualizing the MBs targeted to the disease biomarkers, and has the potential to be a more sensitive and earlier predictor of rupture risk.^{22,36}

A USMI strategy, referred to as modulated acoustic radiation force (mARF)-based imaging, has been developed and validated for the detection and quantification of disease markers both in vitro and in vivo.^{30,37} Acoustic radiation force (ARF) has been demonstrated to increase the binding efficiency of MBs to potential binding sites.^{37,38} The proposed mARF-based imaging method is beneficial when compared with other recently devised radiation force-based techniques due to its capability to achieve quantitative measurement of molecule targets, high specificity, and short protocol times demonstrated in preliminary in vitro and in vivo studies.^{20,30,37}

In the implementation of mARF, first, there is an initial “pre-ARF” phase with only imaging pulses, followed by an “ARF-on” phase where both ARF pushing pulses and imaging pulses are applied. Finally, there is another “post-ARF” phase at the end of the imaging session (Fig. 1A). An empirical model illustrating the binding dynamics of adherent MBs has been described (Fig. 1B).²⁰ Three parameters were extracted from the dynamic response of adherent MBs: the initial signal intensity (SI_{init}), the saturated signal intensity after the ARF application (SI_{sat}), and the residual signal intensity after turning off the ARF (SI_{res}). The residual-to-saturation ratio ($R_{res-sat}$), was defined to represent the adhesion levels of targeted MBs in different groups (Fig. 1C). The presence of the targeted molecular markers on the vascular endothelium,

VEGF receptor 2 (VEGFR-2) in this study is assumed if there is a significant enhancement in the adherent MB signal in the VEGFR-2-targeted MBs group compared with the control group.

In this study, we validated the proposed mARF-based imaging method in vivo using the angiotensin II (Ang II)-induced AAA murine model and VEGFR-2 as the molecular marker. The results indicated that the mARF-based USMI method was able to provide sensitive detection and assessment of AAA in a murine model. We also demonstrated that the measurements obtained using the mARF-based imaging method exhibited a correlation with the level of VEGFR-2 expression during AAA progression.

MATERIALS AND METHODS

Microbubbles Preparation and Conjugation

Biotinylated MBs (mean diameter = 3.2 μm , standard deviation of diameter = 1.8 μm , median diameter = 2.6 μm , and resonance frequency $\approx$ 4.5 MHz³⁹) were prepared using previously described methods.^{40,41} VEGFR-2-targeted MBs were synthesized by conjugating biotinylated anti-mouse VEGFR-2 antibody (clone Avas 12 al; eBioscience, San Diego, CA) to the MB using streptavidin (AnaSpec, Fremont, CA). The control MBs were conjugated to the biotinylated isotype control antibody (clone R35-95; BD Pharmingen, San Diego, CA).

Briefly, the MB aqueous dispersion was first washed 3 times using centrifugal flotation to remove unincorporated lipid micelles. A Multisizer 3 Coulter Counter (Beckman Coulter, Brea, CA) was used to count the number of MBs left after the centrifugal flotation. Streptavidin was added to the MBs at 3 μg per 10 million MBs, then they were incubated for 15 minutes at room temperature. After another centrifugal flotation washing to remove excess streptavidin, MBs were incubated with the antibodies added at a concentration of 1.5 μg per 10 million MBs. The number of antibody molecules per MB was calculated to be approximately 1×10^5 .⁴⁰ The purified MB aqueous dispersion was refrigerated overnight in the cold room under perfluorocarbon headspace. All the conjugated MBs were prepared within 24 hours before use.

Murine Model of AAA

All animal studies were performed under a protocol approved by the Institutional Animal Care and Use Committee at the University of Virginia. C57BL/6 male mice at 8 weeks old (The Jackson Laboratory,

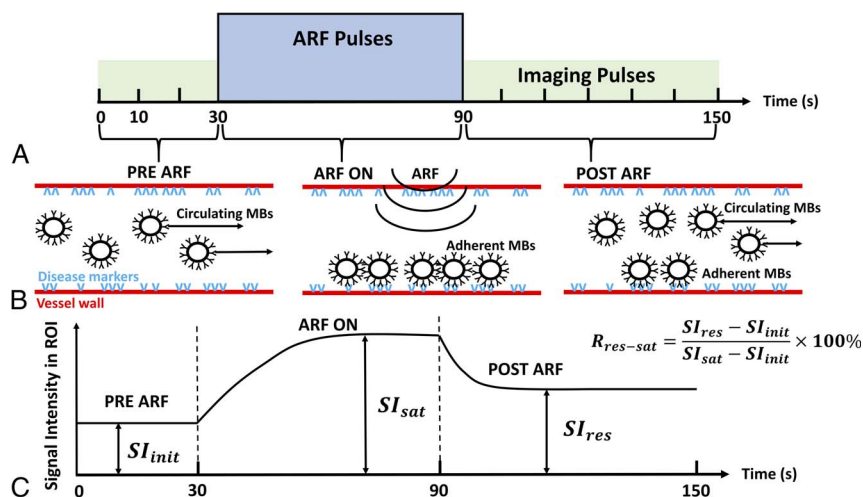


FIGURE 1. The schematic of modulated ARF-based imaging method. A, The mARF-based sequence consists of imaging pulses from beginning to the end and ARF pulses from 30 seconds to 90 seconds. B, Illustrations of dynamic responses of microbubbles at different phases: perfusion during pre-ARF phase; attachment during ARF-on phase; and release of untargeted microbubbles during post-ARF phase. C, The signal intensity curve of adherent microbubbles within the selected region of interest (ROI). Definitions of the parameters are described in the figure.

TABLE 1. The Survival Rate of Mice Before Imaging Sessions in Ang II + BAPN Group and Control Group at Each Time Point

	1 Weeks	2 Weeks	3 Weeks	4 Weeks
Survival rate in Ang II + BAPN group, % (n)	100% (10/10)	50% (5/10)	80% (8/10)	50% (5/10)
Survival rate in control group, % (n)	100% (5/5)	100% (5/5)	100% (5/5)	100% (5/5)

Bar Harbor, ME) were used in this experiment. For each imaging session, 15 mice were divided into 2 groups. The mice AAA model implantation was following the methods described in previous articles.^{42,43} For 10 mice in the experimental group, aneurysms were induced using subcutaneous Ang II (4006473, previously as H-1705; Bachem, Torrance, CA) infusion. The mice received freshly prepared β -aminopropionitrile monofumarate (BAPN) (A3134; Millipore-Sigma, St Louis, MO) solution dissolved in the drinking water (0.2% wt/vol), beginning 2 days before the Ang II pump implantation. Osmotic minipumps (Alzet Model 2004, Cupertino, CA) were filled with Ang II and incubated at 37°C overnight before implanting underneath the dorsal skin of the mice. The control group comprised 5 mice receiving minipumps filled with saline alone without BAPN solution or Ang II. The pumps were set to deliver subcutaneous Ang II or saline at a rate of 1000 ng/kg per minute. The mice in both the Ang II + BAPN group and control group were changed to a saturated fat-enriched diet (42% fat, TD.88137, Harlan Teklad; Envigo, Indianapolis, IN) after implanting the pumps.

Experiment Design

The mice were imaged at 1, 2, 3, and 4 weeks after the osmotic pump implantation. After each imaging session, the aortas were harvested for ex vivo testing. Because of the high rupture rate (approximately 1/3 of the mice) of the Ang II–induced AAA model, the number of survived mice in the Ang II + BAPN group varied at different time points. The survival rate of mice before imaging sessions at each time point in both the Ang II + BAPN group and control group is shown in Table 1. Each mouse received 2 separate bolus injections of MBs. First, the mice received the isotype control antibody conjugated MBs to estimate the nonspecific adhesion “background” signal. After the first imaging session, there was a 10-minute waiting period allowing MBs clearance,⁴⁴ and the VEGFR-2–targeted MBs were injected for the second imaging session.

Ultrasound Imaging Strategy

The mice were imaged using a Vevo 2100 scanner (FUJIFILM VisualSonics, Toronto, Ontario, Canada) with the MS400 probe, operating at a central frequency of 30 MHz. Acoustic radiation force was applied using a separate 5-MHz focused Panametrics transducer (A309S-SU; Olympus, Tokyo, Japan). These 2 transducers were colocalized using a custom-designed clamp to image the aorta and apply ARF to translate MBs. The setup of the instrument is shown in Figure 2. The colocalization was set to be at an imaging depth of 8 mm. The Panametrics transducer was connected to a function generator (AFG3052C; Tektronix, Beaverton, OR) and an amplifier (Model A150; E&I, Rochester, NY). The properties of the generated ARF pulses were described in Table 2, and the ARF pulses with 55 dB amplification were applied to the mice. The Vevo 2100 scanner collected image data continuously for 150 seconds at a frame rate of 13 Hz. The image recording began after the injection of MBs; the first 30 seconds was the “pre-ARF” stage for MBs infusion. Thereafter, the ARF was applied to the mouse for 60 seconds. Followed by the cessation of ARF, there was a 60-second “post-ARF” stage for the release of unbounded MBs. The image data were exported from Vevo 2100 in video data format.

The region of interest (ROI) was selected on the bottom wall of the aneurysmal aortas. To reduce the motion artifacts, we adopted a nonrigid nonparametric diffeomorphic image registration method based on Thirion’s demons algorithm to preprocess all the images.^{45,46} The ROI selection was based on a previously described method.³⁰ Briefly, the entire AAA region was first manually selected on the preprocessed images

(Fig. S1A, Supplemental Material, <http://links.lww.com/RLI/A850>). To further reduce the physiological motion on the vessel wall, all frames within the AAA region were summed together, and a searching window (2 × 0.8 mm) along the bottom vessel wall was used to establish the potential ROI (Figs. S1B, C, <http://links.lww.com/RLI/A850>). Finally, to locate the vessel wall segments with MBs attachment as precisely as possible, the vessel wall within the searching window that had a signal magnitude higher than approximately 40% of the maximum signal magnitude on the bottom vessel wall was defined as the ROI (Figs. S1D, E, Supplemental Material, <http://links.lww.com/RLI/A850>). The magnitude of the signal within the ROI was detected and displayed with time. The SI_{init} , SI_{sat} , and SI_{res} were calculated by averaging the signal intensities within the ROI in 5 seconds before the ARF was applied, 5 seconds before the cessation of ARF pulses, and at the end of the imaging session, respectively. Because each mouse had different initial vessel wall signal intensity, all signal intensity curves were normalized by dividing all signal intensities during the 150 seconds imaging period by the maximum intensity.

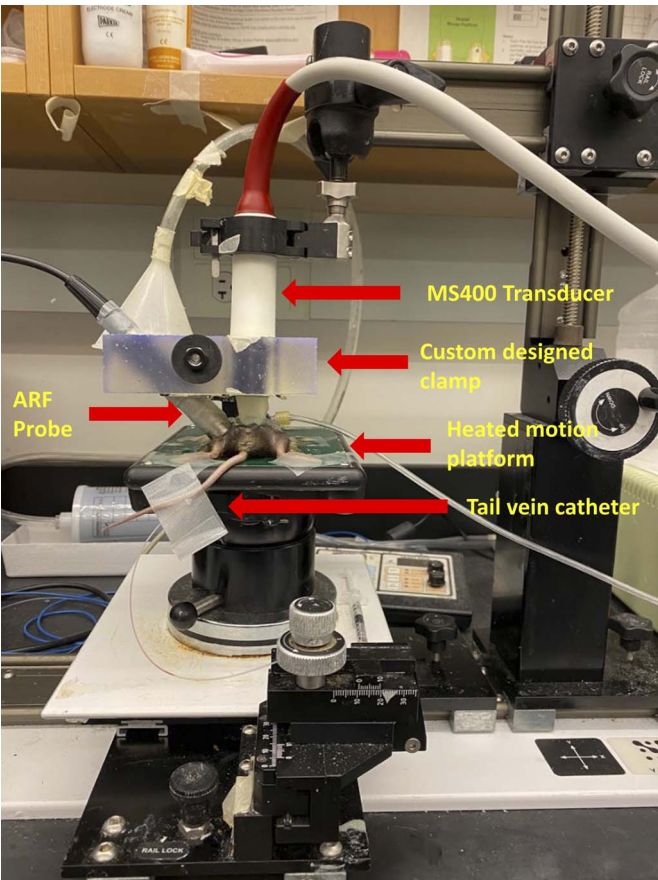


FIGURE 2. Instruments setup. Two transducers were colocalized and secured using a custom-designed clamp. The mouse was placed on a temperature-monitored heated motion platform. The tail vein catheter was performed by a professional veterinary technician.

TABLE 2. The Acoustic Parameters of ARF Pulses

Transducer Type	A309S-SU, Olympus-Focused Panametrics Transducer
Frequency, MHz	5.0
Focal length, in	0.75
Pulse length, cycle	40
Pulse repetition frequency, Hz	1×10^3
Peak-negative pressure, MPa	0.18
Mechanical index	0.08

In Vivo Imaging of Mice

Each mouse was initially anesthetized in the induction chamber using 1.5–2% vol/vol isoflurane (Henry Schein, Dublin, OH). Then, the mouse was placed on a temperature-monitored heated motion stage (TM150, Indus Instruments, Webster, TX) in the supine position while its nose was flushed using a cone connected to 1.0% vol/vol isoflurane outflow. The hair from the mouse abdomen was removed using an electric shaver and depilatory cream. The ultrasonic gel was used to couple the transducer to the mouse and locate the abdominal aorta. The location and the orientation of the transducer were carefully adjusted to align the abdominal aorta within the imaging plane. The image depth lay in the range of 8–12 mm depending on the anatomy of the specific mouse. For each injection of MBs, approximately 20 million MBs diluted in 50 μ L sterile saline were injected via the tail vein catheter using a syringe pump (PHD ULTRA, Harvard Apparatus, Holliston, MA) at the infusion rate of 30 μ L/min (approximately 2.0×10^5 MBs/s).

Ex Vivo Testing

At each time point, after the imaging sessions, the aortas from the mice in both the Ang II + BAPN group and control group were harvested for formalin fixation and paraffin embedding. A subset of mice from each group was used for immunohistochemical analysis of VEGFR-2 expression in aneurysmal and nonaneurysmal segments of the aorta. Immunostaining of VEGFR-2 was conducted based on methodologies modified from prior reports.⁴⁷ Briefly, 4- μ m paraffin sections of healthy versus aneurysmal aortas were deparaffinized, followed by antigen retrieval in a specialized decloaking chamber (NxGen DC2012; Biocare Medical, Pacheco, CA). The slides were then incubated with primary antibody (#9698; Cell Signaling Technology, Danvers, MA) for 12 hours and rinsed with PBS at least 3 times. The slides were then incubated with an anti-rabbit secondary antibody conjugated with Alexa Fluor 594 (A-11037; Invitrogen, Carlsbad, CA) for 1.5–2 hours. After rinsing with PBS, the tissue sections were then counter-stained with DAPI-containing mounting medium (VECTASHIELD, H-1200; Vector Laboratories, Newark, CA). VEGFR-2 staining images were acquired with EVOS M7000 Automated Imaging System (M7000; Thermo Fisher Scientific, Waltham, MA).

Statistical Analysis

Welch unequal variances *t* test (Welch *t* test) and one-way analysis of variance (ANOVA) test were used in this study. The Welch *t* test was able to compare the $R_{res-sat}$ between the Ang II + BAPN group and control group at each time point, and the ANOVA test could reveal whether there is a statistical difference in the $R_{res-sat}$ between different time points. Statistical analysis was performed using RStudio (RStudio, PBC, Boston, MA). All data in the figures were expressed as mean $\pm$ standard deviation.

RESULTS

B-Mode Imaging Results

For each mouse, the ultrasound imaging lasted for 150 seconds, and approximately 2000 frames were collected for each targeted or

control MBs injection. The data were saved in Vevo 2100 after the imaging session and then were exported to the lab workstation in video format. All the data were processed and analyzed using MATLAB (Mathworks, Natick, MA). Figure 3 illustrates the B-mode images of the mice in both the Ang II + BAPN group and the control group after 1-week to 4-week AAA induction procedure before any MBs injection. The images showed that from week 2 there was an increase in the aorta diameter in the Ang II + BAPN group, which indicated that the AAA lesions were induced successfully. In week 3 and week 4, there was a clear aneurysm region in the abdominal aorta of the mice in the Ang II + BAPN group, which had a larger aorta diameter and more asymmetric shape compared with the normal abdominal aorta in the control group.

Figure 4 illustrates the accumulation of both targeted and nontargeted MBs on the vessel wall during the ARF-on stage, as well as the detachment of nontargeted MBs and the remaining targeted MBs on the vessel wall at the post-ARF stage. The images were collected from one mouse in the Ang II + BAPN group and one mouse in the control group at the 4-week time point. The frames at *t* = 90 seconds in both groups showed a signal enhancement on the vessel wall within the ROI compared with the frames at *t* = 0 seconds. Parts of the vessel wall became brighter on the images, which indicated both the targeted and nontargeted MBs were translated to the distal wall by the ARF pulses. The frames at *t* = 150 seconds showed that the aorta from the control group had almost no signal enhancement left 60 seconds after the cessation of ARF, whereas the aneurysm region in the Ang II + BAPN group showed an increase in signal intensity compared with the initial frame collected at *t* = 0 seconds. These images indicated that there was a release of nontargeted MBs from the vessel wall after the cessation of ARF, and the signal increase from the VEGFR-2–targeted MBs was due to targeted molecular adherence to biomarkers on the vessel wall. The signal of the adherent MBs was shown as the “residual signal intensity” on the signal magnitude curve.

Vessel Wall Signal Intensity Enhancement

The mean signal intensity curves of the bottom vessel wall within the ROI of both the Ang II + BAPN group and control group at the 4-week time point are illustrated in Figure 5. The mean signal intensity curves are based on the average signal magnitudes within the ROI of the mice in either Ang II + BAPN group or the control group. The signal intensity curves for each MB injection (VEGFR-2–targeted MBs and isotype control MBs) are also included in Figure 5. Solid lines indicate the mean, and dash lines indicate ± 1 standard deviation.

For the mice in Ang II + BAPN group injected with VEGFR-2–targeted MBs, the signal intensity increased with the application of ARF pulses beginning at *t* = 30 seconds and then decreased to a residual level after the cessation of ARF pulses. For the mice in the control group or the mice in Ang II + BAPN group receiving isotype control MBs, there was an increase in signal intensity after turning on the ARF pulses, and then signal intensity decreased to a lower level that was close to the initial baseline level, which explained why the $R_{res-sat}$ was approximately zero for the mice in the control group or injected with isotype control MBs (Fig. 5A).

The $R_{res-sat}$ for the mice in Ang II + BAPN group receiving VEGFR-2–targeted MBs was 31.8%, significantly higher (*P* < 0.001) than other groups (4.85% for mice in Ang II + BAPN group receiving control MBs, –1.47% for mice in the control group receiving VEGFR-2–targeted MBs, and 3.03% for mice in the control group receiving control MBs) as shown in Figure 5B.

Residual-to-Saturation Ratio at Each Time Point

Figure 6 illustrates the $R_{res-sat}$ at each time point for the mice in both the Ang II + BAPN group and control group after receiving VEGFR-2–targeted MBs. The Welch *t* test revealed that the mean $R_{res-sat}$ of the mice in Ang II + BAPN group exhibited a significant difference compared with the mice in the control group at all time points

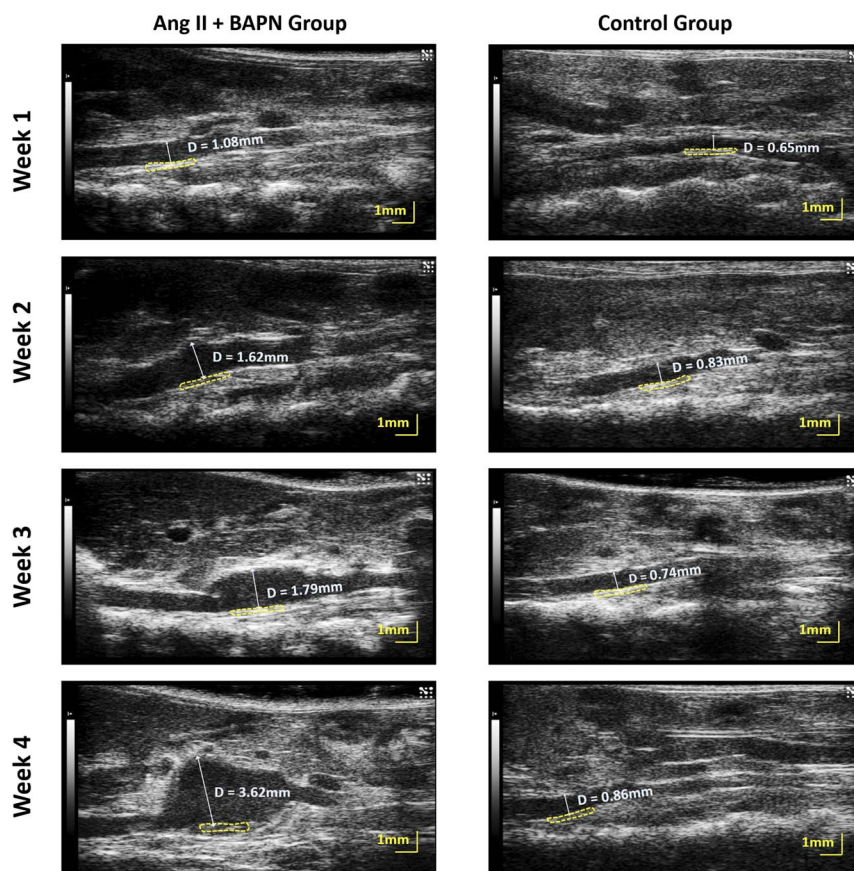


FIGURE 3. The B-mode images of the mice abdominal aortic aneurysm from 1-week to 4-week time points. Images displayed at 50 dB dynamic range. The aneurysmal lesions in Ang II + BAPN group present with enlarged aortic diameters as well as asymmetrical shapes, in comparison to the normal abdominal aortas in the control group. Increased aortic diameters, indicative of the progressive growth of AAA lesions, could be observed through the 1-week to 4-week time points. The yellow dot lines indicated the region of interest (ROI). The diameter of the aneurysm (or aorta) was indicated with arrows.

($P < 0.001$) (Fig. 6A). The mean $R_{res-sat}$ for the mice in the Ang II + BAPN group at 1, 2, 3, and 4 weeks postimplantation were 9.20%, 20.6%, 22.7%, and 31.8%, respectively. Meanwhile, the mean $R_{res-sat}$ for the mice in the control group at 1, 2, 3, and 4 weeks postimplantation were 2.13%, 1.85%, 3.26%, and 4.85%, respectively. These results demonstrated that the mARF-based measurements had the ability to detect the AAA at an early stage of the disease by using targeted MBs that adhered to the vessel wall and the signal intensity of targeted MBs scaled with the expression level of biomarkers associated with the disease.

A one-way ANOVA test revealed that there was a statistically significant difference between the mean $R_{res-sat}$ calculated at 4 time points in the Ang II + BAPN group ($P < 0.005$). A post hoc multiple comparisons test with Bonferroni correction indicated that the mean $R_{res-sat}$ of the mice at 2-week, 3-week, and 4-week time points were significantly higher than the mean $R_{res-sat}$ of the mice at 1-week time point ($P < 0.005$). Although there was no significant difference between the mean $R_{res-sat}$ at 2-week and 3-week ($P = 0.52$), the mean $R_{res-sat}$ of the mice at 4-week showed a significant increase compared with 2-week and 3-week time points ($P < 0.005$). On the other hand, the $R_{res-sat}$ results in the control group at each time point did not show a significant difference under the ANOVA test ($P = 0.47$) (Fig. 6B).

Ex Vivo Testing

Representative images of VEGFR-2 immunofluorescent staining in the abdominal aortas were shown in Figure 7. Representative images from abdominal aortas in each of the Ang II + BAPN group and

control group at each time point were presented. Consistent with prior reports, our data demonstrate that the expression of VEGFR-2 was upregulated in the suprarenal aortic segments from mice challenged with Ang II, in comparison to control mice subjected to saline infusion. Intriguingly, the most prominent upregulation of VEGFR-2 can be observed at the early stage of AAA (week 1 and week 2) rather than the later stage. The possible explanations for no clear difference between the Ang II + BAPN and control group at week 3 and week 4 were included in the Discussion section. Moreover, the increase in VEGFR-2 expression level in week 1 and week 2 correlated with our mARF-based measurements that the $R_{res-sat}$ increased during AAA progression in Ang II-infused mice.

DISCUSSION

In this study, the mARF-based imaging method was demonstrated to be effective as a means of detecting AAA lesions for early diagnosis and assessing the expression level of angiogenesis markers during the AAA progression using VEGFR-2-targeted MBs. The mathematical model developed for the binding and detachment dynamics of targeted MBs in response to modulated ARF was first demonstrated to be consistent with experimental results using a gelatin-made vessel phantom,²⁰ followed by an optical verification study to observe the dynamics of adherent targeted MBs with the application of ARF in blood vessel environments.³⁸ The $R_{res-sat}$ extracted from the mathematical model has also been reported to be an excellent parameter independent of attenuation or imaging path length effects on the acoustic signal magnitude to

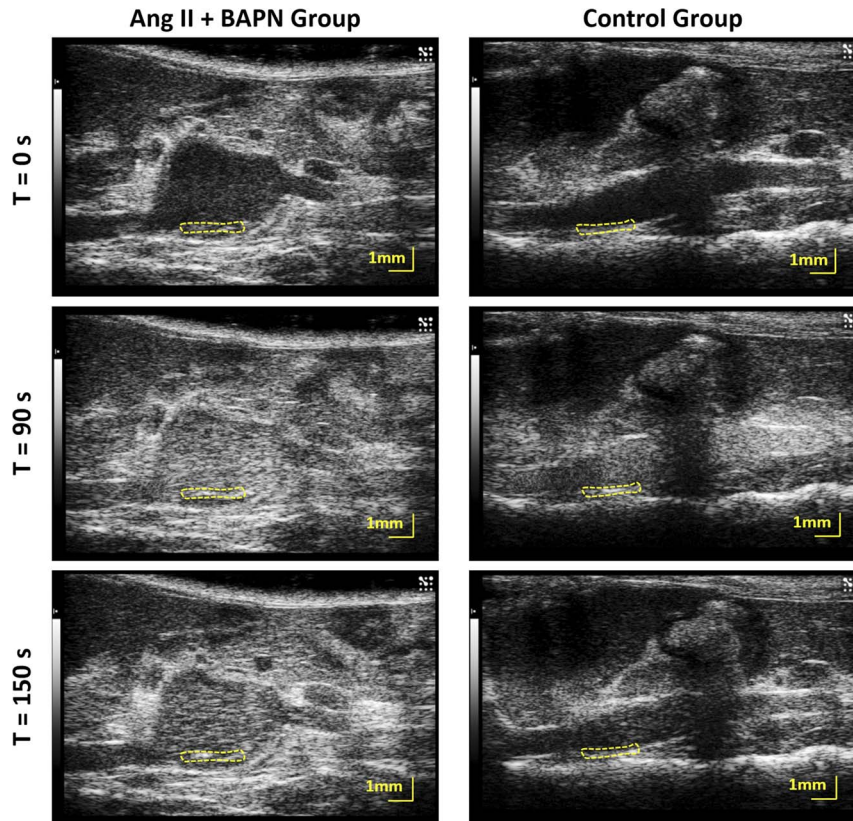


FIGURE 4. The B-mode images of the mouse in either the Ang II + BAPN group or control group at 4-week time point illustrating the change of signal intensity during the imaging session. Images displayed at 50 dB dynamic range. From the top row to the bottom row, the images were selected at $t = 0$ seconds, $t = 90$ seconds, and $t = 150$ seconds, respectively, to represent the signal intensity changes at the pre-ARF phase, ARF-on phase, and post-ARF phase. The yellow dot lines indicated the ROI.

represent the targeted MBs adhesion.³⁷ Compared with some studies that have been working on using targeted MB as a microRNA-based therapeutic agents delivery vehicle for early diagnosis of AAA and prevention of AAA development,⁴⁸ the proposed imaging method allows a more rapid diagnosis by using modulated ARF to translate MBs instead of using destructive pulses to burst MBs and achieves similar high-level performance in the detection of AAA lesions.

The colocalization of the MS400 transducer and a focused panametric transducer was a solution we found for the current in vivo AAA study. In prior mARF-based studies,^{30,37} custom-designed beam sequences were implemented on a programmable ultrasound scanner (Verasonics, Kirkland, WA). However, we found that the programmable scanner with custom-designed sequences did not work well on the mice AAA models as we expected. The quality of the visualization, in terms of spatial resolution and contrast of the aneurysmal region, was insufficient. For example, in several instances, low image resolution restricted us from locating the abdominal aorta and the bottom wall of the aortic aneurysm. After unsuccessful attempts using the programmable transducers, we chose to use a high-frequency and high image-quality commercial ultrasound scanner—Vevo 2100 with the MS400 transducer. The focused transducer and the amplifier were used to ensure that the ARF pulses had similar peak negative pressure and mechanical index as reported in previous experiments³⁷ and had the capability to efficiently translate MBs during the “ARF-on” stage. The colocalized 2 transducers were able to simultaneously image the aneurysmal region with high resolution and push MBs to the vessel wall, which gives us a better visualization of the aortic aneurysm and the signal enhancement of adherent target MBs on the bottom wall of the aneurysm. We observed interference patterns showing up as white vertical stripes during the ARF-on

stage when the ARF single element transducer and the imaging probe were turned on simultaneously. The average intensity of these interference stripes was significantly lower than the signal intensity along the vessel wall. We managed to subtract out these vertical stripes by developing a peak-search and median filtering-based method, and it turned out that there was no meaningful difference in the signal intensity around the aneurysmal region before and after we removed the interference stripes. Therefore, we believe the measurement of SI_{sat} during the ARF-on stage would not be affected by the interference.

The results in Fig. 6A indicated that the residual to saturation ratio $R_{res-sat}$ is an accurate and reliable indicator to detect AAA at early stages based on the VEGFR-2 expression level since all the 4 time points exhibited low P values ($P < 0.001$) between the Ang II + BAPN group and the control group. However, we noticed that the $R_{res-sat}$ measured in this in vivo AAA study is slightly lower than the $R_{res-sat}$ reported in the prior studies^{30,37} described below. The in vitro study reported the mean value of $R_{res-sat}$ could reach approximately 60% in a flow phantom incubated with streptavidin,³⁷ and the in vivo study reported the mean value of $R_{res-sat}$ was 40.9% using an inflammation abdominal aorta model with P-selectin as the targeted molecular marker,³⁰ whereas the highest $R_{res-sat}$ in this study measured at 4-week time point is 31.8%. Although these observations will not have any impact on our conclusion, since the $R_{res-sat}$ of the control group is lower than 5% and significantly different from the Ang II + BAPN group ($P < 0.001$), we believe that the decrease of the $R_{res-sat}$ may reflect how the biomechanical conditions of AAA affect the mARF-based measurements. Compared with the flow phantom and the inflammation model, the geometric changes of the aneurysmal region are unpredictable and different from case to case. The distribution of the molecular biomarkers on the inner

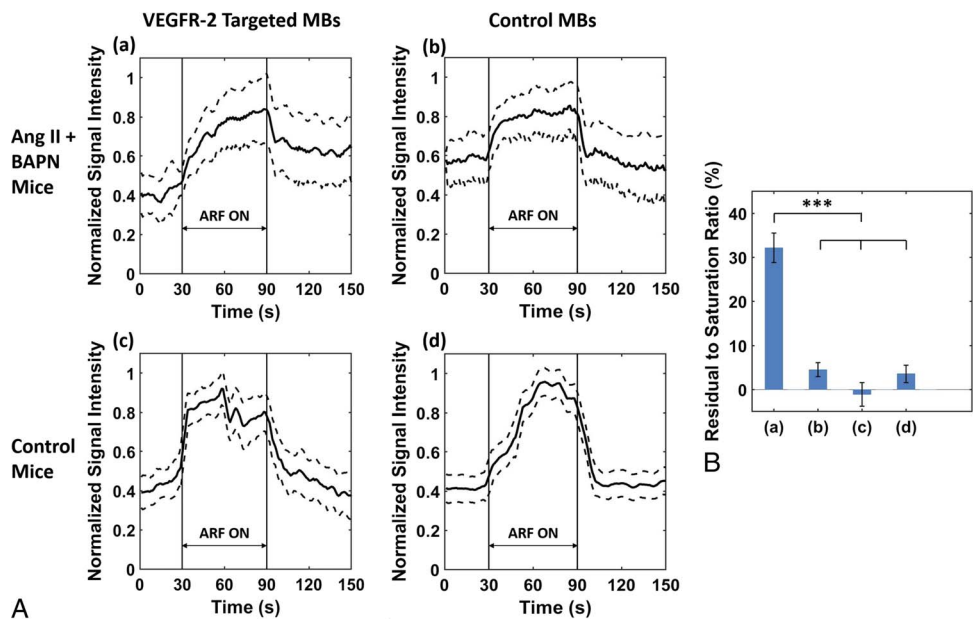


FIGURE 5. The mean signal intensity curves and the residual-to-saturation ratio of the mice in the Ang II + BAPN group and control group at 4-week time point. A, The mean signal intensity curves for the mice in Ang II + BAPN group receiving either targeted or control microbubbles, and for the mice in the control group receiving either targeted or control microbubbles, from panels a to d. B, The residual-to-saturation ratio ($R_{res-sat}$) of the 4 groups showed in (A). Bar charts from panels a to d match the signal intensity curves from panels a to d. *** $P < 0.001$.

surface of the vessel wall was assumed to be symmetric in prior work.³⁰ However, AAA is usually asymmetric, and we were not able to determine which side of the vessel wall had a higher VEGFR-2 expression level. In this study, MBs were only translated to the distal side of the vessel wall relative to the transducer location, and the signal magnitude response on the distal wall was measured, which may lead to an inaccurate value of $R_{res-sat}$. In addition, the complex flow patterns in AAA may make the binding of targeted MBs more difficult compared with the normal aorta. The blood flow patterns in AAA contain one dominant blood flow and low-velocity vortices, which may exhibit as a single vortex, multiple small vortices, or helical vortices.⁴⁹ A prior in silico study reported that the volumetric forces acting on the blood inside the AAA may cause perturbations and vortices to scatter the MBs.⁵⁰ It has been proven that the mARF-based measurement $R_{res-sat}$ is related to the flow velocity in prior in vitro experiment,³⁷ but the influence of vortices remained unknown.

It is possible that these vortices could prevent the MBs from attaching to the vessel wall and might lead to the loss of signal intensity of adherent targeted MBs.

The combination of Ang II and BAPN in the mouse AAA model has been recently introduced to better recapitulate the progressive disease course of AAA in clinical settings. The addition of BAPN, in lieu of a $Ldlr^{-/-}$ or $ApoE^{-/-}$ transgenic background, leads to the irreversible crosslinking of elastin and collagens, and hence aortic stiffening, an aging-associated pathologic feature widely believed as a prelude to AAA. Indeed, multiple studies reported a higher AAA formation rate than the conventional Ang II model, in which the aortic aneurysm and dissection tend to form in the ascending and thoracic aortas instead of abdominal regions.⁵¹ However, as is the case with all AAA models featuring Ang II infusion, some early mortality—typically caused by aortic dissection and rupture—is inevitable; the addition of BAPN to the Ang II-challenged

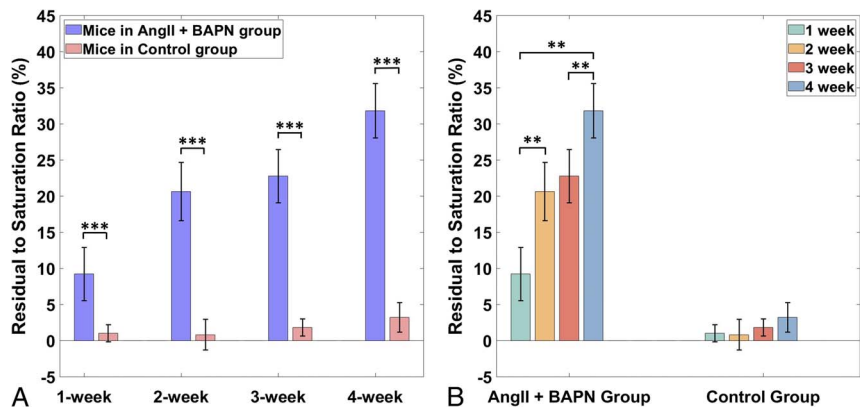


FIGURE 6. The residual-to-saturation ratio ($R_{res-sat}$) at each time point. A, Welch t test revealed that the mean $R_{res-sat}$ of the mice in the Ang II + BAPN group were significantly higher than the mean $R_{res-sat}$ of the mice in the control group at different time points ($P < 0.001$). B, The mean $R_{res-sat}$ of the mice in the Ang II + BAPN group had a significant difference among the 4 time points under 1-way ANOVA test ($P < 0.005$), but not the control group. ** $P < 0.005$; *** $P < 0.001$.

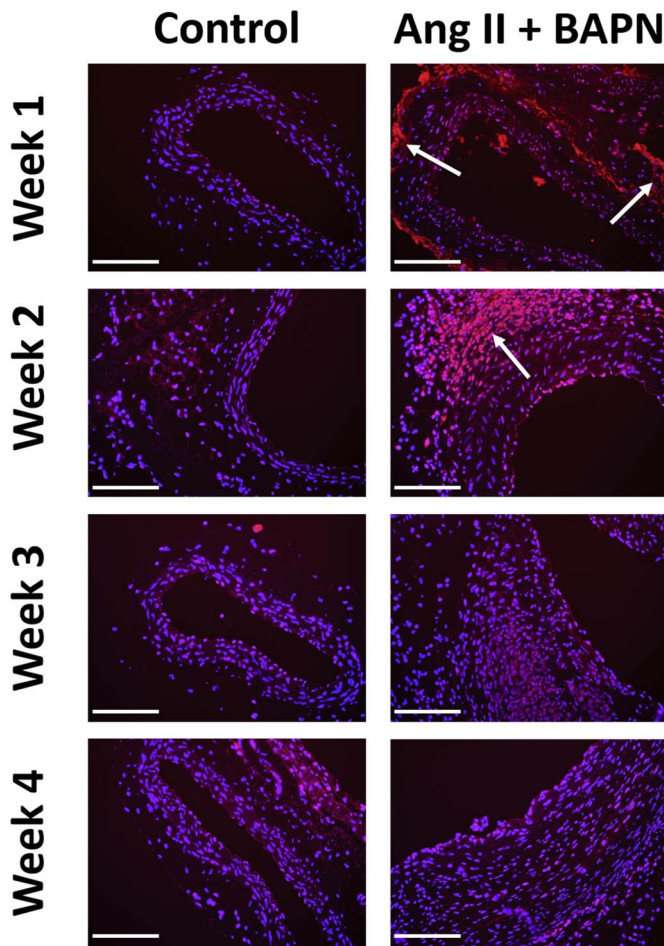


FIGURE 7. The temporal and spatial patterns of VEGFR-2 expressions in AAA (with Ang II infusion) versus healthy aortas (with saline control). Increased expression of VEGFR-2 (as shown in red fluorescent signals, indicated by white arrows) could be observed in the suprarenal aortic segments, particularly in the adventitial and medial layers, from Ang II-challenged mice versus saline control. Scale bar = 125 μ m.

mice further fuels a high rupture rate (31%) in the advanced stages of AAA.⁵² In the current study, the sample size for the AAA group was adjusted accordingly, based on the anticipated high mortality rate, as shown in Table 1. Using this new model, we were able to demonstrate the concomitant increase in abdominal aortic diameters as well as local VEGFR-2 expression. Surprisingly, in immunostaining results, the prominent transmural upregulation of VEGFR-2 was observed in mice at the 1-week and 2-week time points after Ang II infusion, whereas the AAA lesions from later time points only showed a marginal increase in VEGFR-2 expression. Prior reports, despite sparse, showed a persistent upregulation of VEGFR-2 in the medial layers of aneurysmal aortas in mice.^{34,35,53} It is worth noting that these studies used distinct models (the conventional Ang II model or the porcine elastase infusion model), which could account for the temporal and/or spatial differences in the observed VEGFR-2 upregulations. Another possible explanation would be that the aortic cells with the strongest VEGFR-2 expression may not colocalize in the aortic segments/areas with the most advanced inflammation and lamellar disruption, as discussed and detailed in the work by Tedesco et al.³⁵ Besides immunostaining, other quantitative measurements could potentially be applied in future work, such as enzyme-linked immunosorbent assay, to produce a concentration curve for the targeted

molecular marker, which might provide a more intuitive validation of our conclusion that the mARF-based measurement $R_{res-sat}$ is proportional to the targeted biomarker concentration.

Clinical translation of the proposed mARF-based imaging technique still requires careful exploration and consideration. A prior study demonstrated that clinical AAA disease and mouse AAA model had remarkable genomic parallels and similar metabolic responses appeared during the AAA growth,⁵⁴ which may provide us with the opportunity to transfer the AAA detection method based on the expression level of VEGFR-2 or other molecular biomarkers associated with AAA progression to human AAA disease. However, the application of the mARF-based imaging method in clinical AAA detection and assessment will still be challenging. The image depth of the human abdomen may require higher energy ARF pulses to induce waves and push MBs, which could lead to concerns about acoustic safety. Contrast agents used in USMI are well developed in clinic and have an excellent safety profile¹⁸; however, it might take more time and higher cost to synthesize and test the targeted contrast agents for human AAA disease compared with other AAA scanning methods such as computed tomography angiography. In addition to the safety and cost issues, the morphological differences and the physiological motion artifacts may cause problems in observing the signal intensity of adherent targeted MBs in human AAA. It is crucial to develop new image processing methods with more advanced algorithms to eliminate motion artifacts and improve image quality if the mARF-based imaging method could be used in future clinical diagnosis.

In our expectation, the mARF-based USMI method could be used as a semiquantitative measurement, and its main value would be in measuring the molecular biomarker concentration in the same patient over time to assist with the detection and assessment of AAA progression. As we discussed in the previous paragraphs, we realized that the measurement metric $R_{res-sat}$ in this method is a relative value that could be affected by several factors such as blood flow patterns and aneurysm morphology that varied from patient to patient. It may not be general enough to serve as a standardized single measurement of AAA across different patients such as computed tomography angiography. However, it could be used as a patient-specific monitoring method in clinical, working like other semiquantitative monitoring techniques (eg, liver elastography⁵⁵) to benefit the diagnosis and treatment in the staging of the disease and long-term health care.

CONCLUSIONS

In conclusion, we have validated the mARF-based ultrasound imaging technique in vivo using the murine AAA model. After the prior in vitro phantom study and in vivo inflammation study, this is the first time that the proposed mARF-based method has been applied and validated in the mouse AAA model, which has more complex conditions in the lumen and requires a more advanced imaging strategy compared with the previous work. Our results indicated that the $R_{res-sat}$ of the adherent targeted MBs in the mice with AAA lesions was significantly higher than the mice in the control group ($P < 0.001$) from all time points, and the $R_{res-sat}$ measurements also showed a correlation with the VEGFR-2 expression level in the aneurysmal region. Our results demonstrated that the mARF-based method was able to detect the AAA accurately and reliably for early diagnosis by using targeted MBs to assess the expression of angiogenesis biomarkers in AAA lesions. Subject to the successful completion of an extensive effort to translate underlying methods to the human environment, the approach may have the potential to detect and assess AAA risk in clinical settings at the early stages of human AAA disease.

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